ASIM ASLAH P M

AI/ML & Generative AI Developer | AI Research | Full-Stack Software Engineer

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

M.Tech candidate in Artificial Intelligence and Data Science with industry experience building AI-enabled web applications. Develops Python and FastAPI services, integrates pretrained PyTorch models, and ships tested React applications. Current research explores conformal privacy-risk routing for retrieval-augmented generation under PII distribution shift.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Generative AI and Research: RAG research, privacy-risk estimation, conformal risk control research, distribution shift, NLP, named entity recognition

Machine Learning: PyTorch, computer vision, object detection, semantic segmentation, TripoSR, YOLOv8, BiSeNet, OpenCV, NumPy, Pandas

Backend and Data: FastAPI, REST APIs, Uvicorn, Supabase, Cloudflare D1

Frontend and Cloud: React, Next.js, HTML, CSS, Liquid, Shopify, Cloudflare Workers

Testing and Tools: pytest, Node.js test runner, GitHub Actions, Git, GitHub, ESLint, Vite

PROFESSIONAL EXPERIENCE

Software Development Intern, Realviz LLP (Brickrat) | Malappuram, Kerala | Jun 2026 - Jul 2026

- Built the application workflow around pretrained TripoSR, converting one furniture image into downloadable OBJ and GLB assets.
- Developed FastAPI endpoints for upload, preprocessing, inference, validation, preview, and download, extending an initial Gradio prototype into a complete web application.
- Integrated Supabase to persist generated-model metadata and public asset URLs for storage and retrieval.
- Implemented browser 3D previews, PWA behavior, Apple Quick Look USDZ delivery, and Android Scene Viewer paths.

SELECTED PROJECTS

DataVeil | Python, FastAPI, Presidio, spaCy, YOLOv8, OpenCV, BiSeNet, pytest

- Built a local-first application that masks sensitive text, documents, and images before external generative AI use, then restores protected content from user-held recovery data.
- Combined Presidio, spaCy, YOLOv8, OpenCV, BiSeNet, eight masking effects, ORB/ECC restoration, bounded privacy controls, and a 62-test GitHub Actions suite.

ASIMON Fashion Studio | Shopify, Liquid, JavaScript, HTML, CSS

- Developed a completed Shopify storefront with merchant-editable sections, responsive catalog layouts, variant-ready product pages, native cart behavior, and accessible motion controls.

Asim Portfolio | Next.js, React, TypeScript, Cloudflare Workers, D1

- Built and deployed this tested Next.js portfolio with case studies, BYTE guide, accessible controls, Cloudflare D1 feedback, and SEO metadata.

RESEARCH EXPERIENCE

PIIShift-RAG | Ongoing M.Tech Research

- Investigating privacy-risk routing for retrieval-augmented generation when PII patterns shift between development and deployment.
- The proposed method combines PII detection, risk estimation, conformal calibration, and policy-aware routing; implementation and empirical evaluation remain ongoing.

EDUCATION

Master of Technology in Artificial Intelligence and Data Science | Government Engineering College Palakkad | 2025 - 2027

APJ Abdul Kalam Technological University | Machine learning, computer vision, pattern recognition, statistics, and optimization

Bachelor of Technology in Computer Science and Engineering | MES College of Engineering | 2020 - 2024

APJ Abdul Kalam Technological University